

Hridya Vakkalanka

Texas, USA • hridya.vkk@gmail.com • (945) 215-4562
(Open to Relocate)

Experienced data engineering professional with over 4 years of experience in finance and healthcare. Proven ability to build scalable data pipelines, integrate large datasets, and support real-time analytics across cloud platforms. Proficient in Python, SQL, Apache Spark, Kafka, and AWS services including Redshift, Glue, and Lambda. Skilled in implementing secure, compliant data solutions and delivering insights through Power BI and Tableau. Seeking opportunities to drive data architecture initiatives and contribute to enterprise-scale analytics solutions, while collaborating cross-functionally and adapting quickly in dynamic environments.

PROFESSIONAL EXPERIENCE

GOLDMAN SACHS

TX, US

Data Engineer

Aug 2023 - Present

Engineered and optimized enterprise-scale financial data pipelines and analytics solutions to drive predictive modeling, regulatory compliance, and strategic insights across global investment functions

- Led a strategic initiative at Goldman Sachs to optimize and streamline financial data pipelines supporting predictive models for investment risk assessment, resulting in enhanced operational efficiencies across wealth management, asset servicing, and securities divisions.
- Developed and fine-tuned complex **SQL queries and Python scripts** for data cleaning, transformation, and analysis, ensuring consistent, high-quality datasets for quantitative modeling, portfolio risk evaluation, and structured finance analytics.
- Created and implemented **Power BI dashboards to visualize KPIs** across investment performance, trade execution efficiency, and portfolio diversification, enabling actionable insights and a **20%** uplift in reporting turnaround time for internal stakeholders.
- Built and automated **ETL pipelines using Apache NiFi**, enabling real-time ingestion of structured/unstructured data from trading desks, risk systems, and client reporting tools, reducing manual intervention by **50%** and accelerating insight delivery across global investment functions.
- Deployed **Azure Databricks** notebooks for parallel processing of high-volume market and portfolio datasets, using Spark SQL and ML flow for financial signal extraction, risk scoring, and alpha strategy testing.
- Managed and queried large-scale datasets using **Apache Hive** to support complex financial analytics across distributed systems.
- Constructed **data marts** for credit risk, profitability, and investment analytics, enhancing data discoverability for front-office use.
- Ensured strict adherence to **financial data governance** and compliance protocols, including **SEC, MiFID II**, and internal data protection standards, safeguarding client information and ensuring audit readiness for all investment data processes.

HCL TECH

India

Data Engineer

Jan 2019 - Aug 2021

Designed and implemented scalable data engineering and analytics solutions to streamline healthcare operations, deliver actionable patient care insights, and enable real-time clinical decision-making.

- Developed **interactive Tableau dashboards** to deliver real-time insights on patient volumes, outcomes, and risk metrics, improving clinical decision-making and operational efficiency.
- Implemented **Snowflake** as a cloud data warehouse, improving query performance by **20%** and reducing storage costs by **15%** across healthcare datasets.
- Built and deployed **ML models** for fraud detection in healthcare claims using **Logistic Regression and Random Forest**, enhancing security and reducing unnecessary costs.

- Designed **ETL pipelines using Python (Pandas, NumPy, SQLAlchemy)** to extract, transform, and centralize data from EHRs, imaging systems, and patient demographics.
- Employed **Apache Spark Streaming** for real-time processing of patient monitoring data and treatment plans, enabling immediate alerts for critical health events.
- Leveraged **Amazon Athena** to run ad-hoc queries on healthcare data stored in S3, enabling efficient on-demand clinical reporting.
- Utilized **HDFS and MapReduce** to process large-scale patient and clinical data, improving system uptime by **30%** and reducing data loss risk by **40%**.
- Managed unstructured data in **MongoDB** to store patient records and treatment details, simplifying complex healthcare data management.

SKILLS

- **Methodologies:** SDLC, Agile, Waterfall
- **Programming Languages:** Python, SQL, R
- **Packages:** NumPy, Pandas, Matplotlib, SciPy, Scikit-learn, TensorFlow, Seaborn
- **Visualization Tools:** Tableau, Power BI, Advanced Excel (Pivot Tables, VLOOKUP), Amazon QuickSight
- **IDEs:** Visual Studio Code, PyCharm, Jupyter Notebook
- **Cloud Platforms:** AWS (IAM, S3, VPC, EC2, Athena, Glue, Lambda, Step Functions, EMR, Redshift, SageMaker), Google Cloud Platform
- **Databases:** MySQL, Oracle SQL, PL/SQL, PostgreSQL, MongoDB
- **Data Engineering Concepts:** Apache Spark, Apache Hadoop, Apache Kafka, Apache Beam, ETL/ELT, PySQL, PySpark
- **Other Technical Skills:** Data Lake, SSIS, SSRS, SAS, Docker, Kubernetes, Jenkins, Terraform, Informatica, Talend, Snowflake, Google BigQuery, Data Quality and Governance, Machine Learning Algorithms, Natural Language Processing, Big Data, Advanced Analytics, Statistical Methods, Data Mining, Data Visualization, Data Warehousing, Data Transformation
- **Soft Skills:** Cross-functional collaboration, Analytical thinking, Problem-solving, Adaptability, Clear communication, Attention to detail
- **Version Control Tools:** Git, GitHub
- **Operating Systems:** Windows, Linux, Mac OS

CERTIFICATIONS

- AWS Academy Graduate - AWS Academy Cloud Security Foundations
- IBM Data Science Professional Certificate (Coursera)
- DataCamp Certificates in Data Analysis and Python
- SSIS Developer Certification (Udemy)

EDUCATION

ANDERSON UNIVERSITY
Master of Science in Cybersecurity Management

South Carolina, US

ST. FRANCIS COLLEGE FOR WOMEN
Bachelor of Commerce in International Business

Telangana, India